



UC-7429

**Integrated M.C.A. (Sem.-II)**  
**Examination, June-2023**

**IMCA-204**

**Mathematics-II**

**Time allowed : Three Hours**

**Max. Marks : 70**

**Min. Pass Marks : 28**

*This question paper contains three sections as under :*

**Section-A****Max. Marks-10**

This section shall have 01 compulsory question comprising 10 questions (maximum 20 words answer) taking two questions from each unit. Each question shall be of one mark and total marks of this section will be 10. This section will be compulsory in the paper.

**Section-B****Max. Marks-25**

This section will carry 25 marks with equally divided into five long answer type questions (answer about in 250 words) and students are instructed to attempt five questions by selecting one question from each unit.

**Section-C****Max. Marks-35**

This section will carry 35 marks with five long answer type questions comprising one compulsory question (question no. 7) of 15 marks and four questions of 10 marks each. Students are instructed to attempt total three questions with one compulsory question (answer about in 500 words) and any two more questions (answer about in 400 words) out of remaining four questions.

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[Contd...

**SECTION - A**

- 1 (i) Define order of an element of a group.
- (ii) Define cyclic group.
- (iii) Define coset.
- (iv) Define automorphism on group.
- (v) Define integral domain.
- (vi) Define prime field.
- (vii) If  $u = e^{xyz}$ , then find  $\frac{\partial u}{\partial y}$ .
- (viii) State chain rule.
- (ix) Define maxima of function.
- (x) Write conditions for  $f(x, y)$  to have minima.

**SECTION - B****UNIT - I**

- 2 ✓ Prove that the set of  $n$ ,  $n^{\text{th}}$  roots of unity is a multiplicative finite abelian group.

**OR**

Show that every permutation can be expressed as the product of disjoint cycles.

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[Contd...

## UNIT - II

- 3 Show that a non void subset  $H$  of a group  $G$  is a subgroup if and only if

$$a \in H, b \in H \Rightarrow ab^{-1} \in H.$$

OR

For any group  $G$ , prove that its centre  $z(G)$  is a subgroup of  $G$  where

$$Z(G) = \{x \in G \mid xg = gx \forall g \in G\}.$$

## UNIT - III

- 4 Show that every field is an integral domain.

OR

Prove that the set  $S = \{a + b\sqrt{2} \mid a, b \in \mathbb{Q}\}$  is a subfield of the field  $(\mathbb{R}, +, \times)$  of real numbers.

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[Contd...

## UNIT - IV

- 5 If  $u = \log(y \sin x + x \sin y)$ , then verify :

$$\frac{\partial^2 y}{\partial x \partial y} = \frac{\partial^2 y}{\partial y \partial x}.$$

OR

If  $u = \tan^{-1}\left(\frac{x^3 + y^3}{x + y}\right)$ , then prove that :

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \sin 2u.$$

## UNIT - V

- 6 Find the points where the function  $x^3 + y^3 - 3axy$  has maximum or minimum values.

OR

Find the maxima or minima of

$$\sin x + \sin y + \sin(x + y).$$

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[Contd...

### SECTION - C

- 7 (a) For any element of a group  $G$  prove that

$$o(a) = o(x^{-1}ax) \quad \forall x \in G.$$

- (b) Show that every subgroup of a cyclic group is also cyclic.

- 8 Show that the order of every subgroup of a finite group is a divisor of the order of the group.

- 9 Show that every prime field of characteristic zero is isomorphic to the field of  $\mathbb{Q}$  of rational numbers.

- 10 If  $u$  is homogeneous function of degree  $n$  of variables  $x, y, z$ ; then prove that :

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = nu.$$

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[Contd...

- 11 Find the maximum and minimum value of ;

$$u = a^2x^2 + b^2y^2 + c^2z^2$$

where  $x^2 + y^2 + z^2 = 1$  and  $lx + my + nz = 0$ .

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